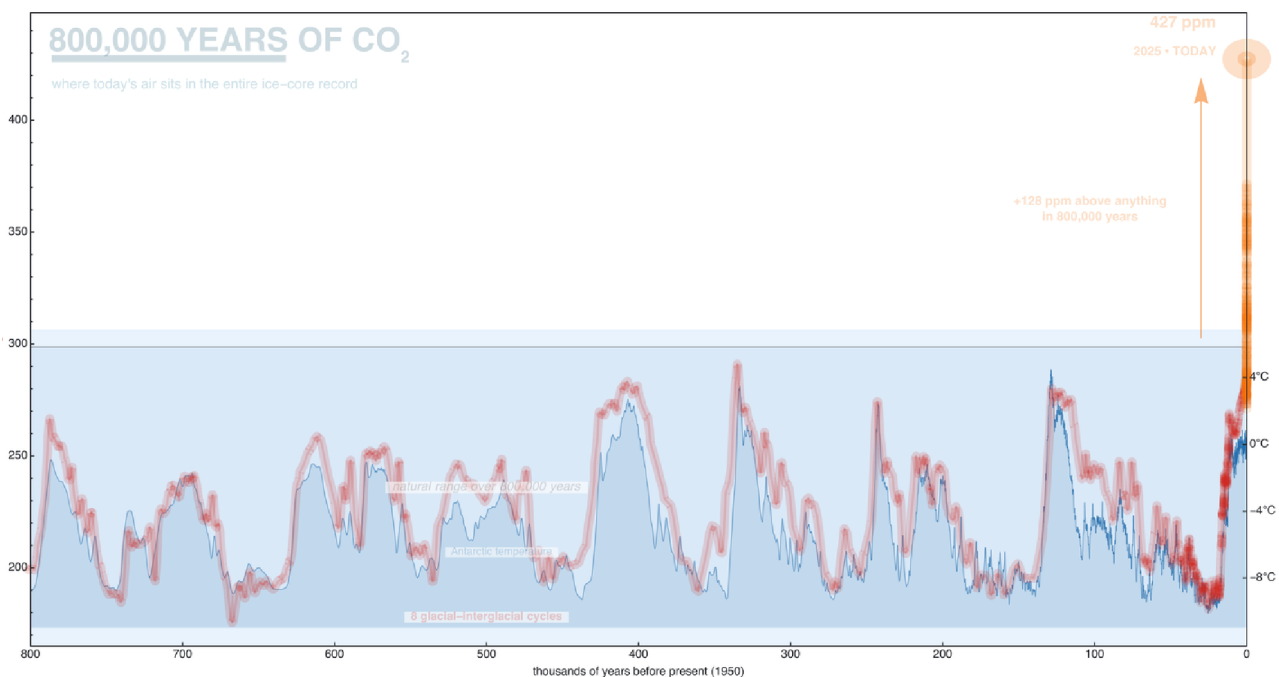


800,000 Years of Climate

CO₂, temperature, and the rhythm of the ice ages — read from the EPICA Dome C ice core in the Wolfram Language

A single column of Antarctic ice holds a continuous 800,000-year diary of Earth's atmosphere. Bubbles trapped in the ice preserve ancient air; the ice itself records temperature through its hydrogen isotopes. From two published EPICA Dome C records we reconstruct the **eight glacial-interglacial cycles**, expose their **~100,000-year Milankovitch rhythm** with a power spectrum, rebuild the **orbital pacemakers** from first principles, ask **which moves first** (CO₂ or temperature), and follow the rhythm back **five million years** through the Mid-Pleistocene Transition. Today's **427 ppm of CO₂** sits about **128 ppm above anything** in the 800,000-year record — reached in a century and a half, not ten thousand years — and was last matched some **3 million years ago**.



Eight hundred millennia of CO₂ (red) and Antarctic temperature (blue silhouette). The grey band is the natural range over the whole record; the orange spike at right is the modern instrumental rise to today.

1. The frozen archive

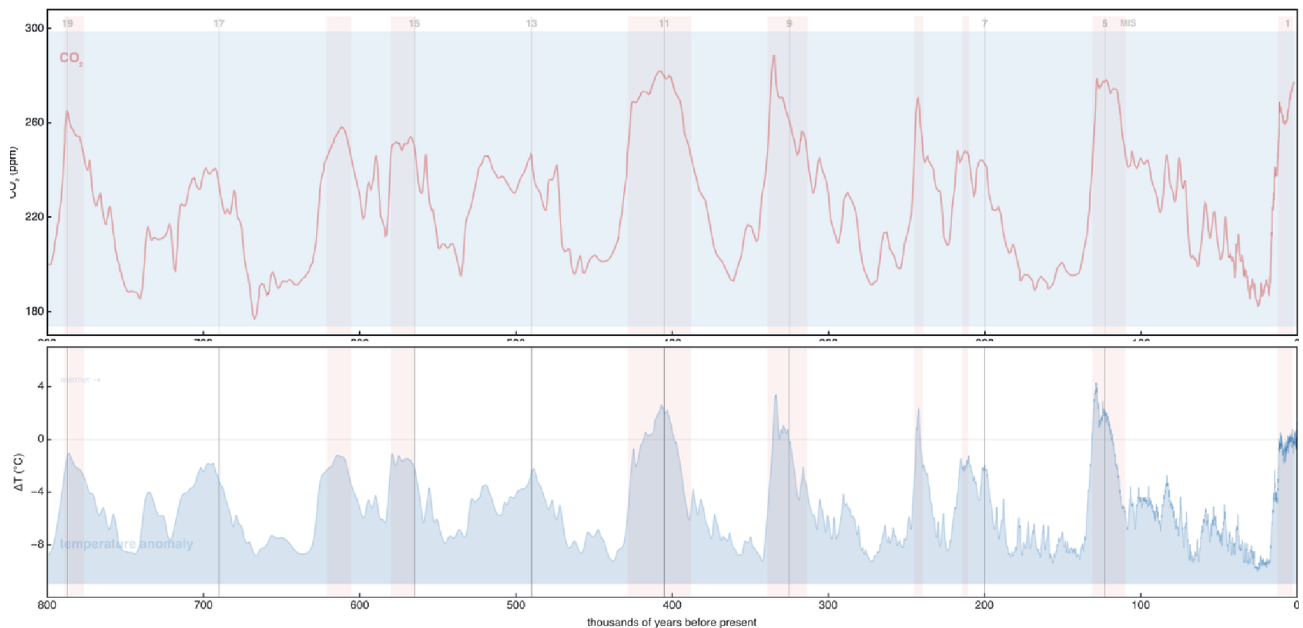
Snow that falls on the East Antarctic plateau rarely melts. It compresses, layer on layer, into ice up to three kilometres thick. The European Project for Ice Coring in Antarctica (EPICA) drilled a core at **Dome C** (75°S, 123°E) that reaches back **~800,000 years** — eight complete ice-age cycles. Two quantities are read from it:

- **Atmospheric CO₂** from air bubbles sealed in the ice. As snow turns to ice, it traps a sample of the contemporary atmosphere. We use the 800 kyr Antarctic composite of [Bereiter et al. \(2015\)](#).
- **Temperature** from the **deuterium** (heavy hydrogen, δD) content of the ice. Colder air delivers isotopically lighter snow, so δD is a thermometer. We use the reconstruction of [Jouzel et al. \(2007\)](#) on the EDC3 age scale.

⚠ **Read the temperature honestly.** This is a **deuterium-derived Antarctic temperature anomaly** — not $\delta^{18}O$, and **not a global mean**. It tracks the glacial–interglacial structure superbly but is a regional, isotope-based estimate relative to the mean of the last 1,000 years. Also, CO₂ is dated by the age of the *trapped gas*, whereas temperature is dated by the age of the *ice*; the offset between them (Δ age) means overlaying the two curves is an excellent **visual** comparison, not a precise phasing analysis. "Before present" means before 1950.

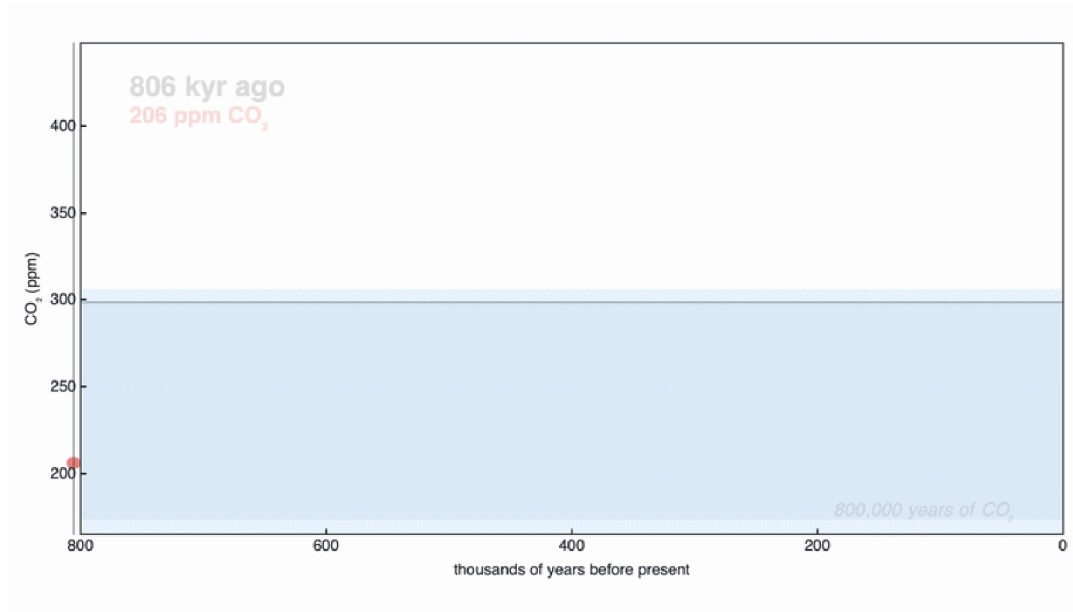
2. Eight glacial–interglacial cycles

Plotted across the full 800,000 years, both records march in a striking sawtooth: a slow, ragged slide into each glacial maximum followed by an abrupt **termination** into a warm interglacial. The interglacials are the odd-numbered marine isotope stages (MIS) — our own Holocene is MIS 1, the last interglacial (the Eemian, ~123 kyr ago) is MIS 5.



CO₂ (top) and Antarctic temperature (bottom) over 800 kyr, present at right. Dotted guides mark the interglacial peaks (MIS stages, after the LR04 chronology); the grey band is the natural CO₂ range. The two curves rise and fall almost in lockstep.

Watch the record assemble itself, oldest to youngest — and then watch the present arrive:

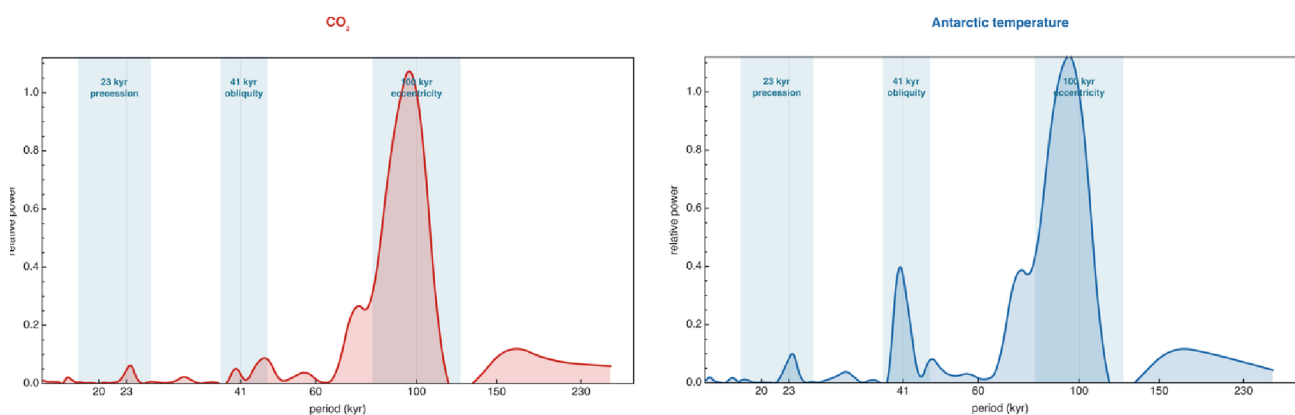


Sweeping through 800,000 years. The ice-core CO_2 stays inside its band for the whole record; only at the very end does the instrumental curve leave the chart.

3. The 100,000-year rhythm

The cycles are not random. Taking a power spectrum of each record — resampling to an even 2-kyr grid, detrending, Hann-windowing and an FFT — concentrates the power at the **Milankovitch periods** of Earth's orbit:

- **~100 kyr** — eccentricity (the shape of Earth's orbit);
- **~41 kyr** — obliquity (the tilt of its axis);
- **~19–23 kyr** — precession (the wobble of the axis).



Power spectra of CO_2 (left) and Antarctic temperature (right). CO_2 is dominated by the ~100 kyr eccentricity band; temperature carries both the 100 kyr and a strong ~41 kyr obliquity peak, with a precession bump near 23 kyr.

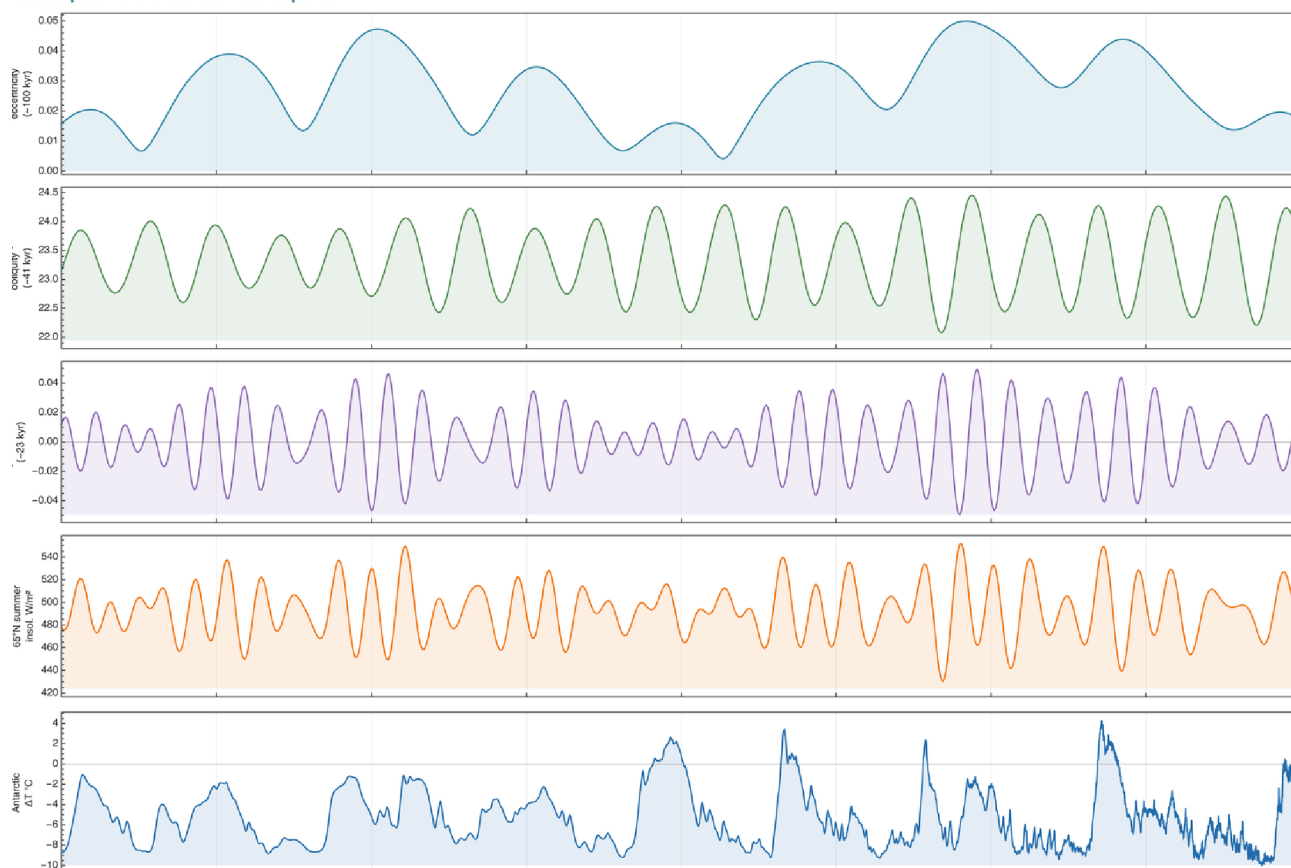
The late-Pleistocene cycles are **paced near 100 kyr**. This is the famous "100 kyr problem": the eccentricity *forcing* is itself weak, so the dominant 100 kyr beat reflects eccentricity-modulated pacing amplified by ice-sheet and carbon-cycle feedbacks rather than direct orbital energy. Temperature's strong 41 kyr peak is the fingerprint of axial tilt. (These are simple illustrative periodograms — enough to reveal the bands, not formal red-noise significance tests.)

4. The orbital pacemakers

We can rebuild the pacemakers themselves. From the [La2004 astronomical solution](#) (Laskar et al. 2004) we take eccentricity, obliquity and the longitude of perihelion, and compute the classic orbital pacing index — the **65°N summer-solstice daily-mean insolation** (Berger 1978), the sunlight available to melt northern ice sheets (a pacing index, not the sole driver — feedbacks and ice-sheet dynamics matter too):

$$Q = \frac{S}{\pi} \left(\frac{a}{r} \right)^2 (H_0 \sin \phi \sin \delta + \cos \phi \cos \delta \sin H_0)$$

orbital pacemakers vs the climate response

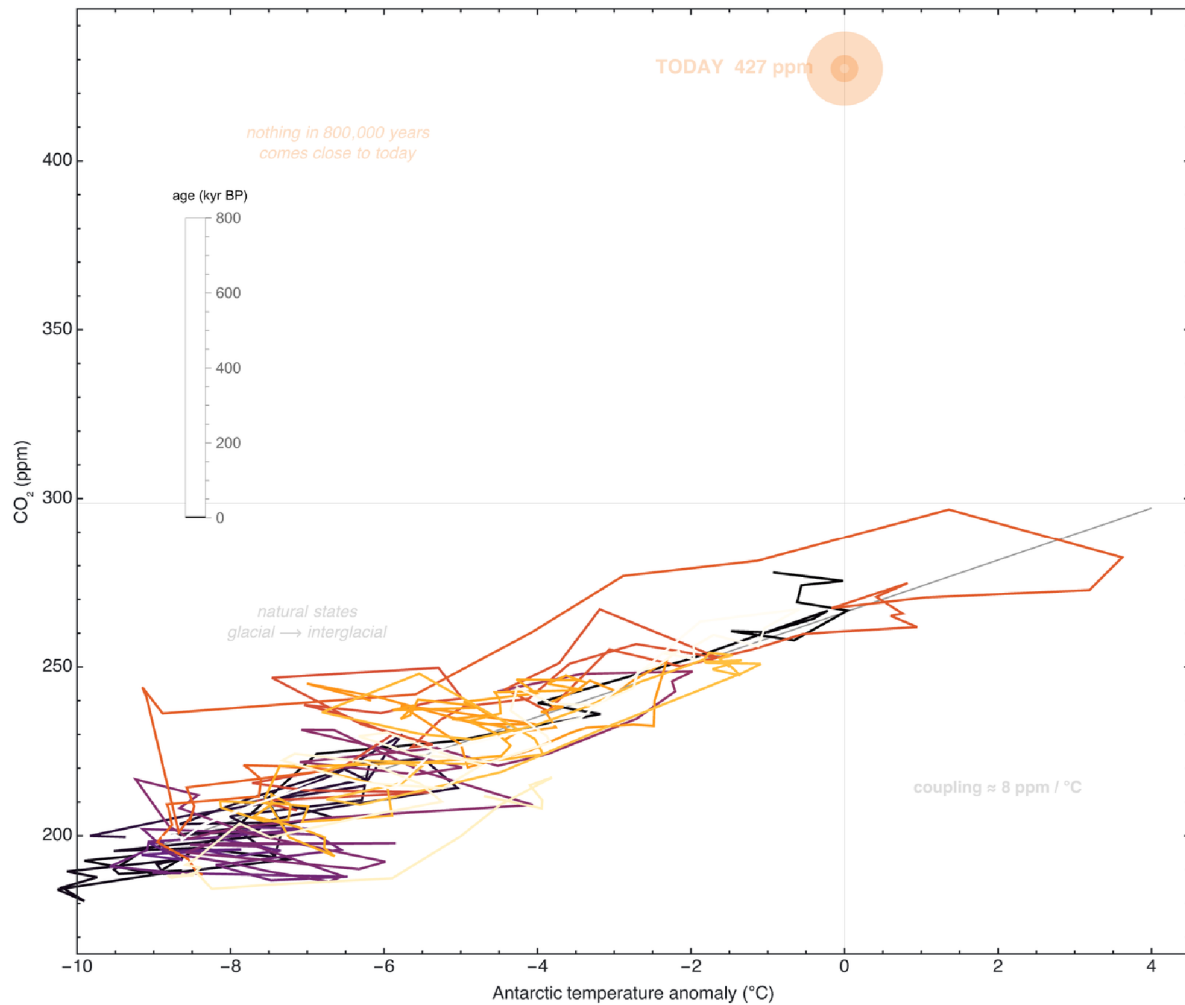


The La2004 orbital parameters — eccentricity (~100 kyr), obliquity (~41 kyr), climatic precession (~23 kyr) — and the computed 65°N summer insolation, above the Antarctic temperature response. Present at right.

△ **Pacing, not a simple cause.** Orbital changes set the *timing* of the glacial cycles, but the temperature swings are far larger than insolation alone can explain. CO₂ acts as a powerful **amplifying feedback**: warming oceans and shifting circulation release CO₂, which warms the planet further. At deglaciations Antarctic temperature often **leads** CO₂ by several centuries — so this is a coupled feedback system, not "CO₂ caused the ice ages" nor the reverse.

5. The tight CO₂–temperature coupling

Plotting CO₂ against temperature, with the 800-kyr trajectory coloured by age, the natural climate states collapse onto a tight diagonal — a linear fit gives roughly **8 ppm of CO₂ per °C** of Antarctic change (a descriptive slope, not a physical sensitivity). Glacial and interglacial worlds are two ends of one well-worn groove.

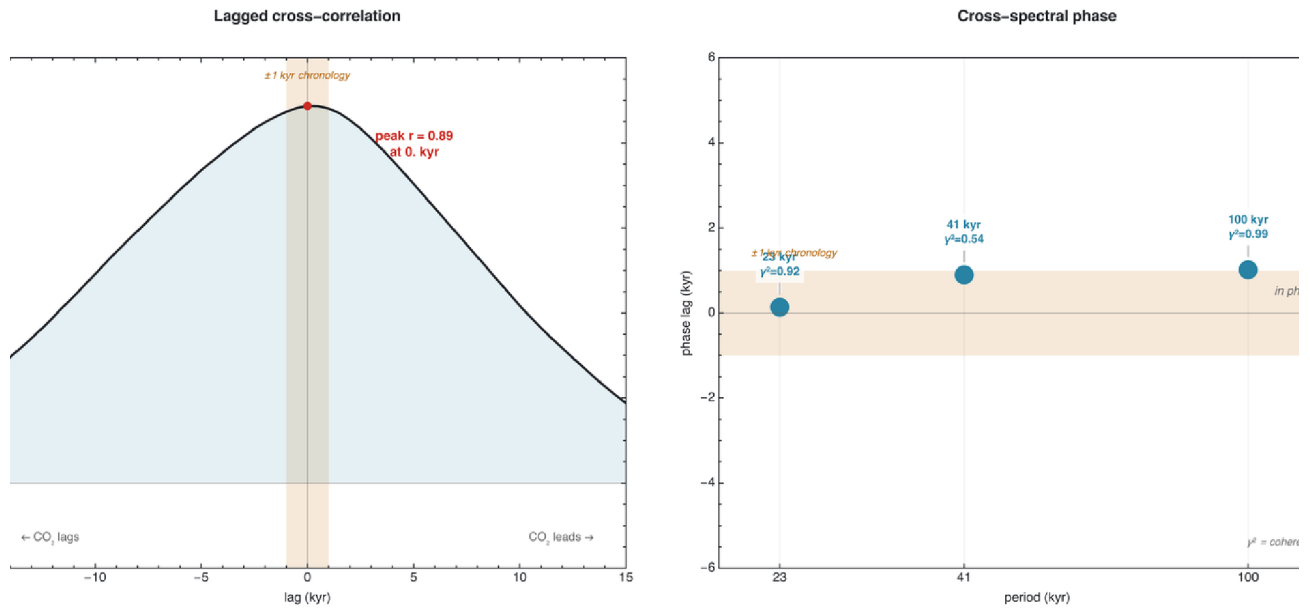


CO₂ versus Antarctic temperature anomaly; the path is coloured from oldest (dark) to youngest. Every one of the last eight cycles lives in the same diagonal cloud — and today's CO₂ sits far above all of it.

And then there is today. The orange point floats high above a phase space that 800,000 years of natural variability never came close to entering.

6. Does CO₂ lead or follow temperature?

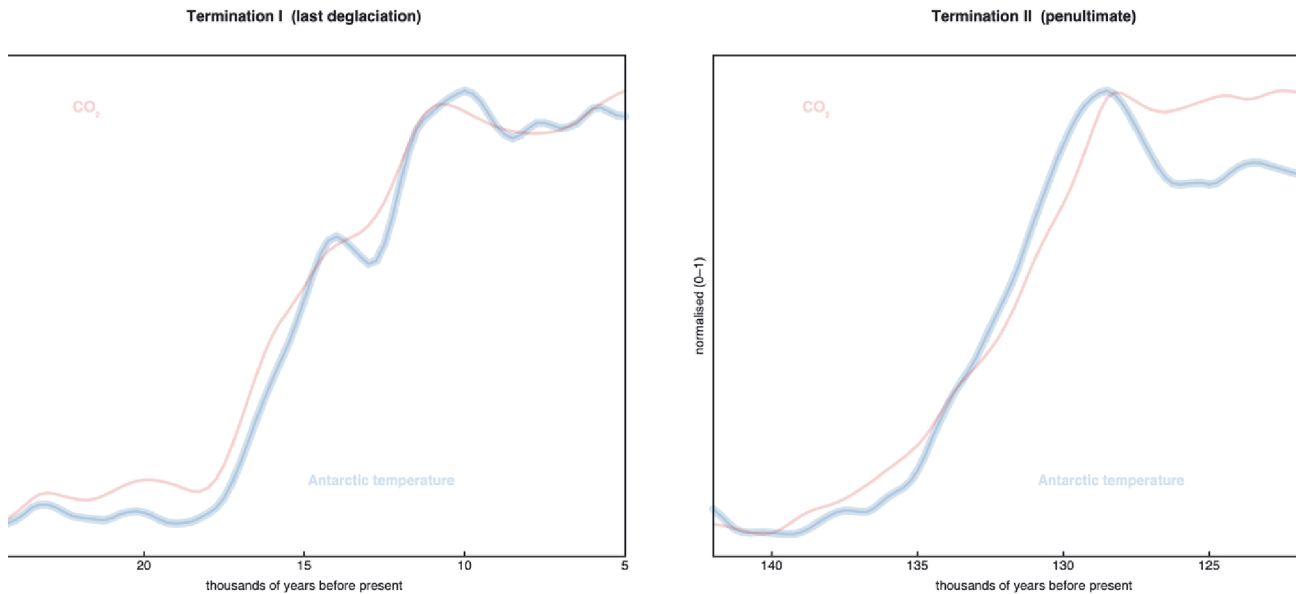
If the two are so tightly coupled, which moves first? Two standard tools answer this — **lagged cross-correlation** in the time domain, and **cross-spectral phase** in the frequency domain. On our resampled records the cross-correlation peaks at $r \approx 0.89$ essentially at **zero lag**, and the phase at every Milankovitch band (100, 41, 23 kyr) sits within **~1 kyr of zero**, with high coherence.



Left: lagged cross-correlation of CO₂ and temperature — a smooth peak at essentially zero lag (positive = CO₂ leads). Right: cross-spectral phase as a lag at each Milankovitch band; all three sit within ~ 1 kyr of zero — the scale of the gas-/ice-age chronology offset — with coherence γ^2 labelled.

⚠ **Why our data can't settle it — and what does.** Our CO₂ sits on a **gas-age** scale and our temperature on an **ice-age** scale (different chronologies, offset by the modelled Δ age of order ~ 1 kyr at Dome C). The apparent ~ 1 kyr lag is comparable to that offset, so these data show tight, in-phase coupling but **cannot resolve the true centennial lead/lag**. That requires CO₂ and temperature from the *same* core on a synchronised timescale. When that is done: Antarctic temperature can *slightly lead* CO₂ at deglaciations (Caillon 2003 found ~ 800 yr; Parrenin 2013 / Pedro 2012 revised it to near-synchrony, 0 ± 200 yr); and during the last deglaciation CO₂ rose **before most of the global-mean warming** (Shakun et al. 2012). The consistent picture: orbital pacing $\rightarrow$ Southern-Ocean warming $\rightarrow$ CO₂ release $\rightarrow$ global amplification. A coupled feedback, with CO₂ the great amplifier.

The clearest way to see it is to zoom into the deglaciations themselves. Overlaying CO₂ and Antarctic temperature (each normalised within the window) across the last two terminations, the two **rise essentially together** — so close that the order of the first few centuries is, in these records, lost inside the Δ age uncertainty rather than plainly visible.

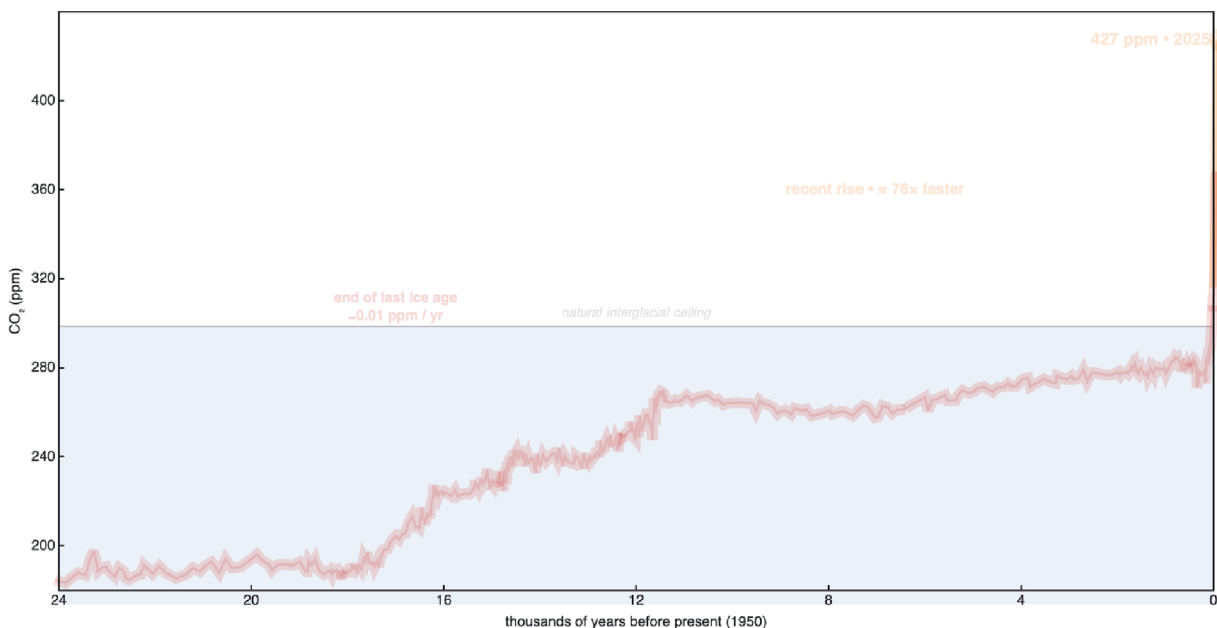


The last two deglaciations, CO₂ (red) and Antarctic temperature (blue) on a common grid with identical smoothing, each normalised within the window, younger at right. A shape comparison (not a timing measurement): they climb almost on top of each other, any residual offset being within the gas-/ice-age chronology offset.

7. The modern breakout

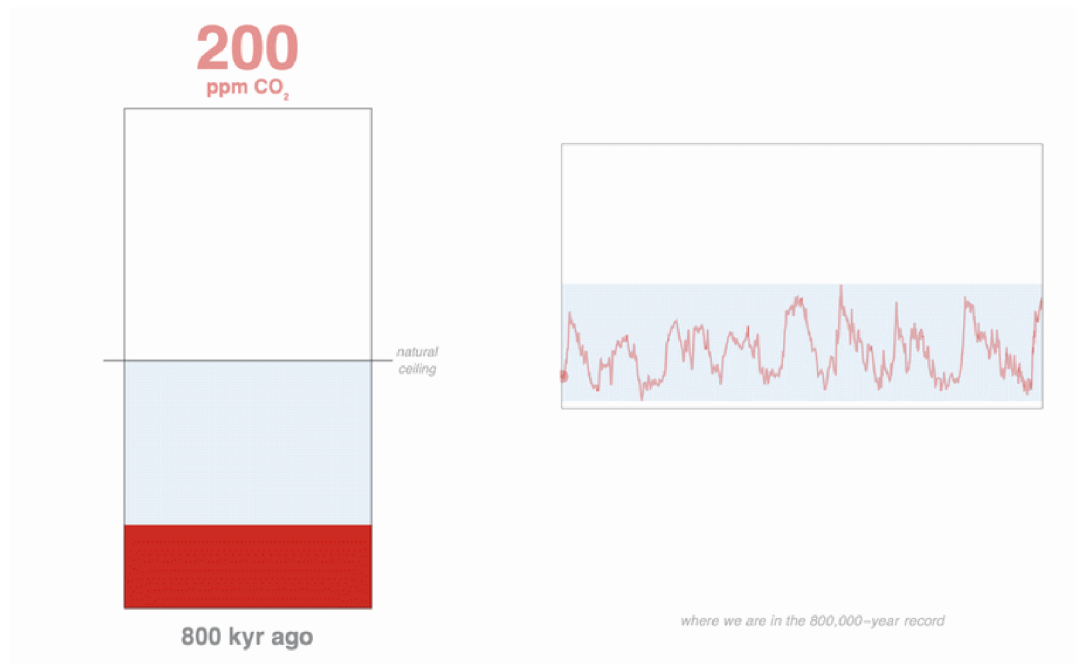
Over the whole ice-core record, CO₂ ranged from about **174 ppm** in the depths of glacials to about **299 ppm** in the warmest interglacials. In 2025 the instrumental Mauna Loa value was **427 ppm** — roughly **128 ppm above** the highest natural level in 800,000 years.

The **rate** is as startling as the level. The natural deglacial rise out of the last ice age — about 90 ppm — took some ten thousand years (~0.01 ppm/yr). The rise since ~1850 has run more than **seventy times faster**.



The last 24,000 years. The gradual deglacial rise of CO₂ (red) gives way to the near-vertical instrumental "blade" (orange) to today's value — off the top of the natural band.

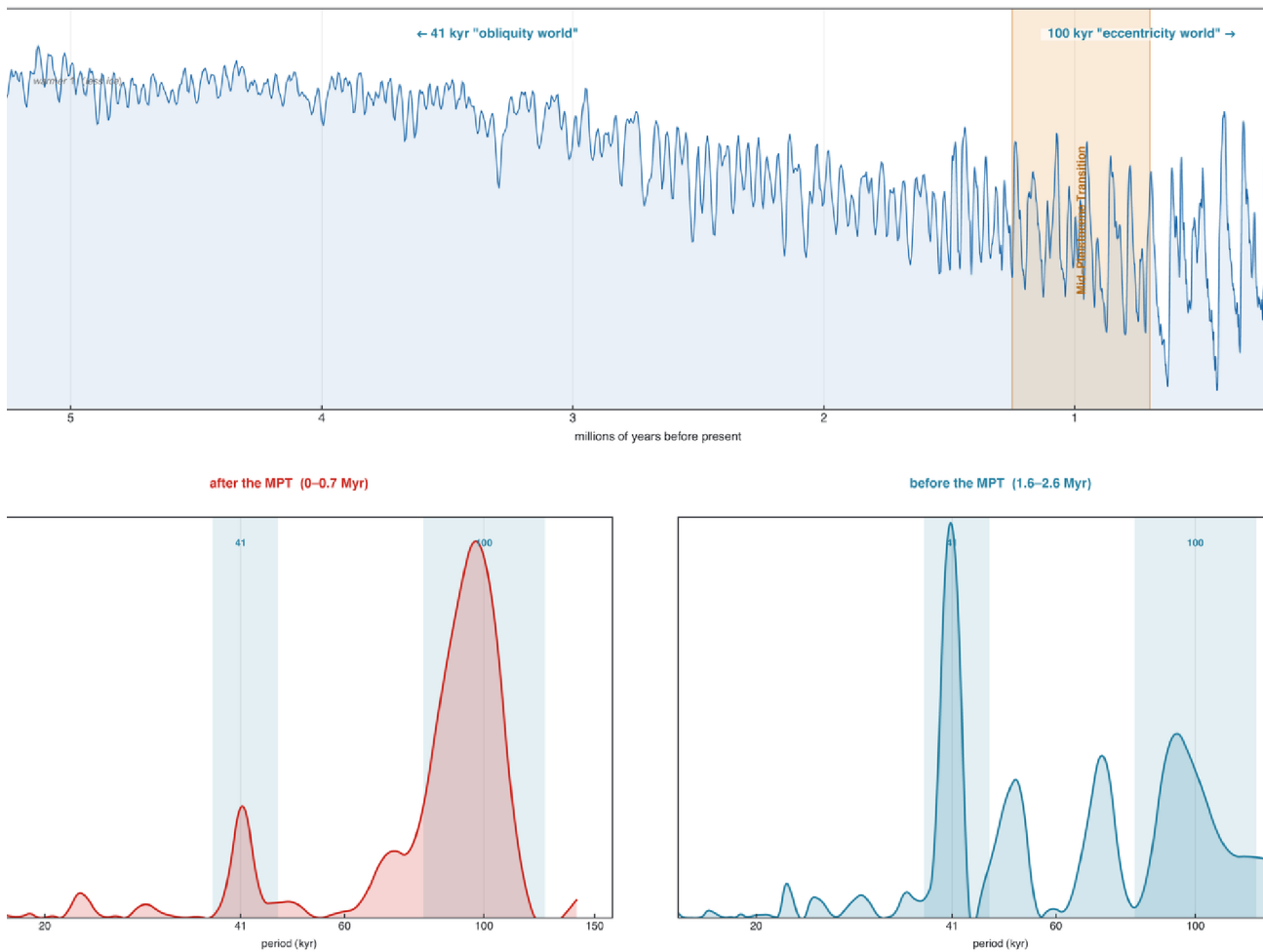
Seen as a gauge filling through time, the picture is visceral: the column tracks the natural band for the entire record, then bursts straight through the ceiling.



A CO₂ gauge swept through 800,000 years. Crimson is the natural range; the orange is the present-day excess that carries CO₂ far past anything in the record.

8. Beyond the ice: five million years

The ice cores stop at 800 kyr, but the **LR04 benthic $\delta^{18}\text{O}$ stack** (Lisiecki & Raymo 2005) — a global proxy for ice volume and deep-ocean temperature — reaches back **5.3 million years**. It exposes a striking change in the climate's rhythm: before about **1 million years ago the world ran on a ~41 kyr (obliquity) beat**; afterwards it switched to the **~100 kyr rhythm** of our eight cycles. This is the **Mid-Pleistocene Transition**.

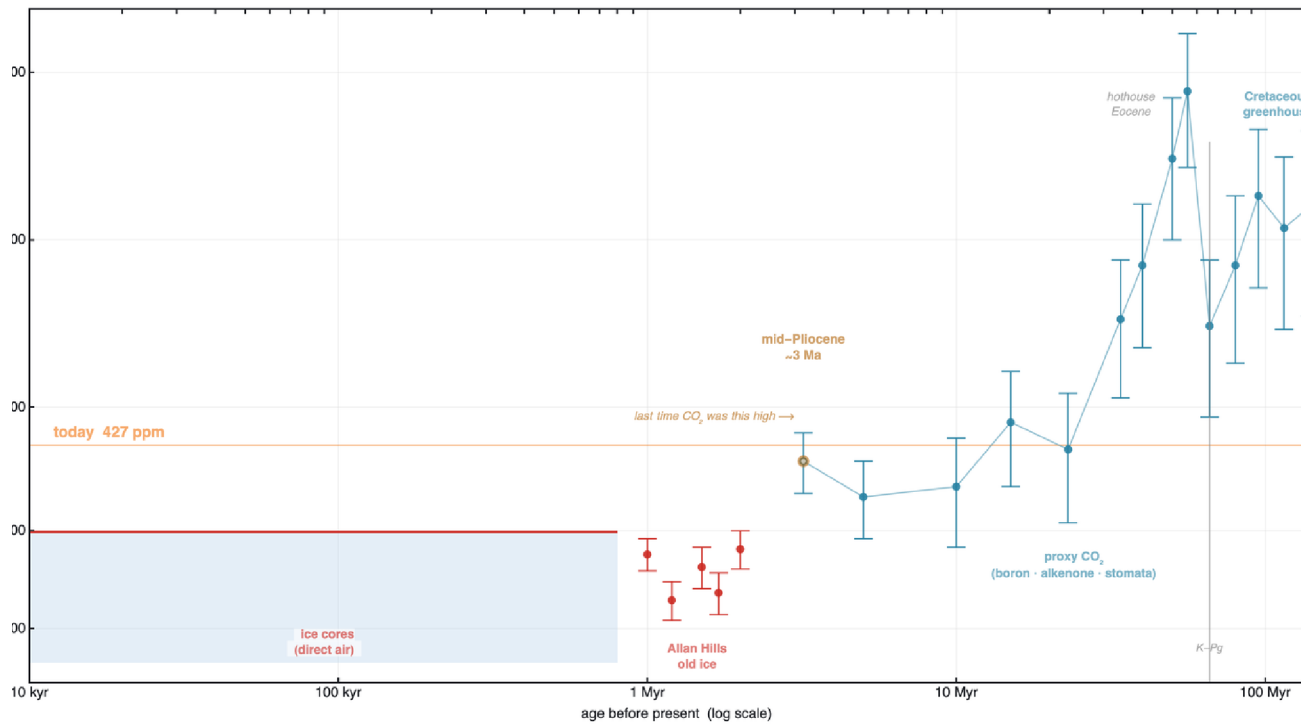


Top: 5.3 Myr of LR04 benthic $\delta^{18}O$ (warmer up), present at right; the cycles grow larger and longer across the Mid-Pleistocene Transition (shaded). Bottom: power spectra in two windows recover the switch — a 41 kyr peak before the transition, a 100 kyr peak after.

Re-running our spectrum in two windows recovers it directly: **before** the transition the dominant period is **~41 kyr**; **after** it is **~100 kyr** — the same beat we measured in the ice cores.

9. Deep time: CO₂ over 150 million years

How unusual is 427 ppm on the longest view? Beyond the ice we rely on discontinuous old ice (**Allan Hills**, ~2 Myr; Yan et al. 2019) and geochemical **proxies** — boron isotopes, alkenones, fossil-leaf stomata — synthesised across the Cenozoic by the **CenCO₂PIP project** (Hönisch et al. 2023) and across the wider Phanerozoic by **Foster, Royer & Lunt (2017)**. On a logarithmic time axis the whole sweep fits in one frame — out to the **Cretaceous greenhouse** ~100 million years ago, when CO₂ ran near 1,000–1,500 ppm.



CO₂ from 10 kyr to 150 Myr on log axes (curated published estimates beyond the ice; see caveat). The narrow ice-core band (blue) and Allan Hills ice (red) stay below ~300 ppm; geochemical proxies (blue, with uncertainties) climb past the Eocene hothouse to the Cretaceous greenhouse. The K-Pg boundary (66 Ma) is marked. Today's level (orange line) was last seen, within proxy uncertainty, in the mid-Pliocene ~3 million years ago.

▲ **The Pliocene benchmark.** The last time CO₂ was comparable to today's 427 ppm (within proxy uncertainty) was the **mid-Pliocene, roughly 3 million years ago** — a world ~2–3°C warmer with seas an estimated 10–20 m higher. Deep-time CO₂ estimates carry **large uncertainties** and are far sparser than the ice cores; the values shown are **representative published estimates, hand-compiled from the cited sources** — not fetched data or a continuous measured record.

10. Conclusions

From Antarctic ice and the records that reach beyond it, the picture is clear:

- CO₂ and Antarctic temperature have cycled together through **eight glacial-interglacial cycles** over 800,000 years, tightly coupled (~8 ppm/°C).
- Those cycles are **paced by Earth's orbit** — the 100 / 41 / 23 kyr Milankovitch bands appear directly in the spectra, and the **Mid-Pleistocene Transition** from a 41 kyr to a 100 kyr world is visible in the 5-Myr LR04 record.
- Lead/lag analysis shows the two are **essentially in phase** within the chronological uncertainty; CO₂ is the great **amplifier** of orbitally-paced change and globally leads temperature.
- Today's CO₂ is **entirely outside the 800,000-year experience** — and was last sustained **~3 million years ago**, in a far warmer, higher-sea world.

11. Data & methods

△ **Scope.** This is a data-visualisation notebook. It shows published, openly-archived records **as their authors released them** — no re-tuning or gap-filling beyond the resampling noted — together with standard, reproducible analyses. It makes **no projections and advocates no position**; the aim is simply to let the measurements be examined directly, and any interpretation is left to the reader.

All pure Wolfram Language. Every record except the deep-time CO₂ panel is **fetches live** with `URLRead`, cached as CSV, and loaded by a shared package; the §9 deep-time CO₂ points are hand-compiled from the cited literature (flagged there). Figures display age as 800 – age so the present is at the right; spectra resample to an even grid before the FFT; lead/lag uses lagged correlation and a smoothed cross-spectrum; insolation follows Berger (1978).

- **CO₂ (800 kyr):** Bereiter et al. (2015), Antarctic composite (NOAA WDS Paleo).
- **Temperature:** Jouzel et al. (2007), EPICA Dome C deuterium temperature, EDC3 scale (NOAA WDS Paleo).
- **Modern CO₂:** NOAA Global Monitoring Laboratory, Mauna Loa & global annual means.
- **Orbital forcing:** Laskar et al. (2004), La2004 solution (IMCCE).
- **Deep-time temperature:** Lisiecki & Raymo (2005), LR04 benthic $\delta^{18}\text{O}$ stack (NOAA WDS Paleo).
- **Deep-time CO₂:** Yan et al. (2019, Allan Hills), CenCO₂PIP (Hönisch et al. 2023) and Foster, Royer & Lunt (2017, Phanerozoic) — curated published estimates (see §9 caveat).

Source, build scripts and this notebook: github.com/mthiel74/IceCores.

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13. Acknowledgements

This notebook rests entirely on measurements generously made public by others. The Dome C records are the work of the **EPICA community** and the investigators cited above; the orbital solution is from the **IMCCE** (Laskar and colleagues); the deep-time compilations are the work of the CenCO₂PIP consortium and the authors cited in §9. Data were obtained from the **NOAA World Data Service for Paleoclimatology** and the **NOAA Global Monitoring Laboratory**, whose long-term archiving makes work like this possible. All analysis, figures and animations are pure Wolfram Language.